We have not been successful in interpreting drift-rates of individual bursts in terms of accelerations because of the scarcity and the uncertainty of individual measurements, and because the velocity calculations are very sensitively dependent on the coronal density model assumed. A significant result was obtained only when all of the available measures were combined.

We conclude, then, that frequency drift-rates of Type II bursts, determined from radio spectral observations, indicate a systematically greater acceleration for the sources of bursts followed by geomagnetic disturbance than for the sources of bursts followed by no disturbance.

M. B. Wood
C. S. Warwick

National Bureau of Standards,
Boulder, Colorado.
Oct. 8.

[1] Thompson, A. R., Paris Symposium on Radio Astronomy, 210, R. N. Bracewell, Ed. (Stanford, 1959).
[2] Maxwell, A., Thompson, A. R., and Garmire, G. (to be published).
[3] Roberts, J. S., Paris Symposium on Radio Astronomy, 194, R. N. Bracewell, Ed. (Stanford, 1959).
[4] Wild, J. P., *Aust. J. Sci. Res.*, A, 399 (1950).
[5] Hartz, T. R., Paris Symposium on Radio Astronomy, 554, R. N. Bracewell, Ed. (Stanford, 1959).
[6] Crow, E., Davis, F., and Maxfield, M., Nav. Ord. Report, 3369 (1955).

## Natural Remanent Magnetization of Pre-Cambrian Gneiss of Ongul Islands in the Antarctic

A number of rock samples of Pre-Cambrian biotite hornblende granodioritic gneiss from the Ongul Islands, where the Japanese Antarctic Research Base Syowa was set up, were collected by one of us (T.N.) in February 1957 and in January 1959. This was for the purpose of determining magnetic polarization of the antarctic rocks in view of recent progress in palæomagnetism, including the problems of polar wandering and continental drift.

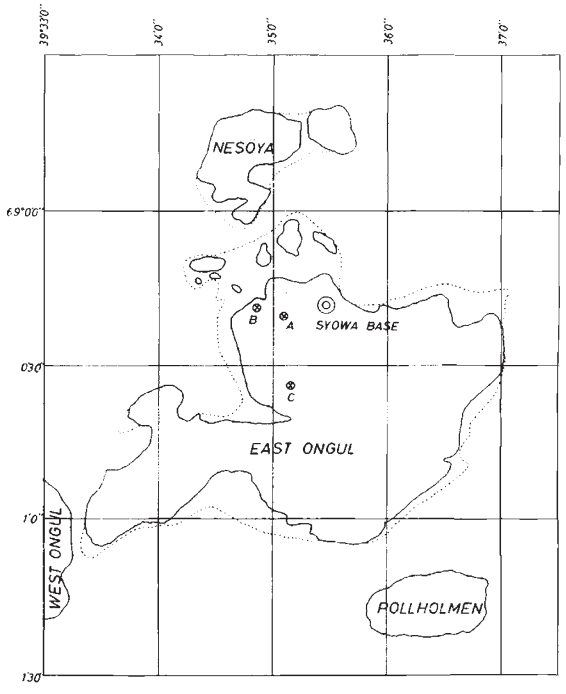



Fig. 1. Localities *A*, *B*, and *C* of sampling Pre-Cambrian gneiss in Ongul Island

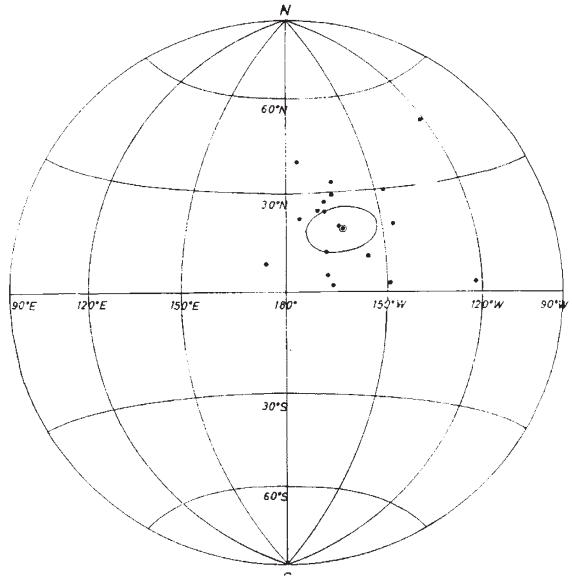



Fig. 2. Position of the north pole of the Earth's centred dipole estimated from magnetic polarization of Antarctic Pre-Cambrian gneiss

The locality of Syowa Base is about 69°00′S. and 39°35′E. Rock samples examined here were collected from three sites in East Ongul Island, as shown in Fig. 1.

These metamorphic rocks *in situ* have a clear banded structure, dip and strike of the gneissosity-planes ranging from 51° to 54° and from N. 20°W. to N. 01°W. respectively.

According to the results of laboratory examination of their magnetic properties, ferromagnetic minerals contained in these rocks are nearly pure magnetites and hematites, having their Curie point at 580°C. and 680°C. respectively, but those responsible for natural remanent magnetization are mainly hematites.

Among a number of collected rock-samples, those which have unstable remanent magnetization are eliminated according to the well known experimental criteria[1] (that is, stability test against d.c. and a.c. magnetic fields and thermal demagnetization).

The results of this test showed that, generally speaking, remanent magnetization of these rocks, intensity of which is very weak compared with the others and is of the order of magnitude of their isothermal remanence, is unstable.

The number of rock specimens whose magnetic polarization was measured was 4 from site *A*, 8 from site *B* and 6 from site *C* (Fig. 1). Intensity and direction of remanent magnetization of these 18 specimens are given in Table 1, and position of the North pole of the earth's centred dipole corresponding to each magnetic polarization is plotted in Fig. 2. The average position of the North pole thus estimated is given as :

lat. = 19°N. long. = 163°W.,

the radius of circle of confidence of 5 per cent being 7°.

At present, the process of formation of the geological structure concerned has not yet been made clear. According to the preliminary results of geological and geomorphological studies of the Ongul Islands[2], however, it seems that the inclination of the planes of banding of gneiss was caused by a certain crustal deformation after completion of metamorphism of those rocks concerned.

Assuming therefore that the gneissosity-planes were horizontal when thermal metamorphism of this

© 1959 Nature Publishing Group

TABLE 1

| Sample No. | Intensity (e.m.u./gm. × $10^5$) | Direction | | Corrected Direction | |
|---|---|---|---|---|---|
| | | Decli. | Incli. | Decli. | Incli. |
| *A*1 | 7·2 | N.33°E. | 78° | N.73°E. | 31° |
| *A*2 | 5·8 | N.20 E. | 40 | N.41 E. | 8 |
| *A*3 | 7·6 | N.16 W. | 50 | N.31 E. | 30 |
| *A*4 | 7·4 | N. 8 W. | 60 | N.44 E. | 30 |
| Pole position derived from mean values of the above four. lat. = 18°N, long. = 137°.5W. Corrected. lat. = 2°.5S, lat. = 96°W. | | | | | |
| *A*1 | 12·2 | N.34°W. | 69° | N.52°E. | 32° |
| *A*2 | 18·6 | N.10 W. | 68 | N.52 E. | 33 |
| *A*3 | 9·0 | N.37 W. | 61 | N.36 E. | 40 |
| *A*4 | 14·4 | N.49 W. | 40 | N. 2 E. | 44 |
| *A*5 | 20·4 | N. 9 W. | 41 | N.25 E. | 22 |
| *A*6 | 22·2 | N.42 W. | 58 | N.38 E. | 46 |
| *A*7 | 13·0 | N.29 W. | 58 | N.33 E. | 40 |
| *A*8 | 21·2 | N.33 W. | 66 | N.46 E. | 42 |
| Pole position derived from mean values of the above eight. lat. = 20°.5N, long. = 166°.5W. Corrected. lat. = 3°.5N, long. = 107°.5W. | | | | | |
| *C*1 | 5·8 | N.27°W. | 39° | N.10°E. | 27° |
| *C*2 | 6·8 | N.29 W. | 42 | N.14 E. | 30 |
| *C*3 | 9·2 | N.35 W. | 64 | N.39 E. | 42 |
| *C*4 | 8·6 | N.50 W. | 70 | N.47 E. | 48 |
| *C*5 | 33·6 | N.35 W. | 62 | N.45 E. | 43 |
| *C*6 | 39·2 | N.31 W. | 50 | N.29 E. | 40 |
| Pole position derived from mean values of the above six. lat. = 16°N, long. = 168°W. Corrected. lat. = 2°.5N, long. = 115°.5W | | | | | |

area took place and consequently thermo-remanent magnetization of those rocks was produced, the direction of magnetic polarization of each specimen is transformed to that referred to the assumed ancient co-ordinates of the horizon and the meridian. The direction thus transformed of each specimen is shown also in Table 1.

The average position of the North pole of the earth's dipole corresponding to the transformed magnetic polarization is given as :

$$\text{lat.} = 3°\text{N. and long.} = 107°\text{W.},$$

where the radius of circle of confidence of 5 per cent is 7°.

It may be concluded from this result that the North pole of the Earth's magnetic dipole was situated approximately at 107°W. in longitude near the equator in the Pre-Cambrian, assuming that there has been no remarkable drift and rotation of the antarctic continent. In the southern hemisphere, palæomagnetic studies on Pre-Cambrian rocks in Australia[3] and South Africa[4] have already been carried out. For comparison, the results of these Australian and South African data are shown in Table 2 together with the present one.

Comparing these results to each other, it will be seen that the pole position derived from the present data is in rather good agreement with that derived from the South African data, while the Australian data, especially the last two results in Table 2, are appreciably different from the upper two. It might be suggested that the rough agreement between the present and the South African results can be attributed to the condition that these two localities are situated at nearly the same longitude though they are separated from each other by about 35° in latitude.

TABLE 2

| Locality | Author | Pole Position | |
|---|---|---|---|
| | | Lat. | Long. |
| Antarctic | Nagata & Shimizu | 3°N. | 107°W. |
| South Africa | Gough (ref. 3) | 8°S. | 137°W. |
| Australia | Irving & Green (ref. 3) | 30°N. | 121°W. |
| | | 51°N. | 18°W. |
| | | 6°N. | 14°W. |

As for the Pre-Cambrian pole position, a number of data have also been reported with respect to European and American rocks in the northern hemisphere. Although these northern hemisphere data[5] do not give an unique pole position of the earth's dipole, it may be said that most of those pole positions are situated between the equator and 35°N.

Summarizing all data obtained hitherto, including the present antarctic one, it may be substantially said that the pole-position of the earth's dipole, estimated by the palæomagnetic method, was situated near the equator in the Pre-Cambrian Period.

T. NAGATA
Y. SHIMIZU

Geophysical Institute,
University of Tokyo.
Aug. 27.

[1] Nagata, T., 'Special Report on Palæomagnetism', Trans. Rome Meeting A.G.A., I.U.G.G., 343.
[2] Yoshikawa, T., and Toya, H., Antarctic Report, No. 1, 1 (1957). Tatsumi, T., and Kikuchi, T., Antarctic Report, No. 7 (1959) (in the press).
[3] Gough, D. I., *Mon. Not. Roy. Astro. Soc., Geophys. Suppl.*, **7**, 196 (1956).
[4] Irving, E., and Green, R., *Geophys. J. Roy. Astro. Soc.*, **1**, 64 (1958).
[5] See, for example, a summary by Irving, E., *Geophys. J. Roy. Astro. Soc.*, **2**, 51 (1959).

## GEOLOGY

### The Geological Time-Scale

SINCE my communication on this subject was written[1], much has happened. Prof. J. L. Kulp[2] has just published a time-scale in which the duration of the geological periods is given as follows : Ordovician, 490–430 m.y. ; Silurian 430–410 m.y. ; Devonian 410–355 m.y. ; Mississippian 355–330 m.y. ; Pennsylvanian 330–275 m.y. ; Permian 275–220 m.y. ; Triassic 220–180 m.y. ; Jurassic 180–135 m.y. ; Cretaceous 135–70 m.y. An almost identical geochronology ($\pm$5–10 m.y.), awaiting publication, was communicated to me privately by Prof. A. Holmes[3] a month ago. The Oxford age of 305 m.y. for Dartmoor granite (by potassium/argon assay on biotite) has been confirmed by Dr. H. Faul[4] of U.S. Geological Survey (292 m.y. and 288 m.y. on the same rock by the same method). A full isotopic assay of Hercynian pitchblende from Geevor, Cornwall, has yielded to Dr. A. G. Darnley[5] of the Geological Survey of Great Britain a comparable date of 288 m.y. From all this it is confirmed that an extension of the Holmes *B* time-scale is called for, and that the best values are, as one would expect, a little higher than the averages of the experimental results published previously[1]. There is however, no support from recent researches for the conclusion that the Lower Cambrian is older than 600 m.y.

C. F. DAVIDSON

Department of Geology,
University of St. Andrews,
Scotland.
Nov. 4.

[1] *Nature*, **184**, 1310 (Oct. 24, 1959).
[2] Kulp, J. L., Program 1959 Annual Meeting, Geol. Soc. America (Nov. 2, 1959).
[3] Holmes, A. (personal communication, Oct. 6, 1959).
[4] Faul, H. (personal communication, Oct. 1959).
[5] Darnley, A. G., Geol. Surv. Great Britain, Age Determination Rep., 6 (Aug., 1959).

© 1959 Nature Publishing Group